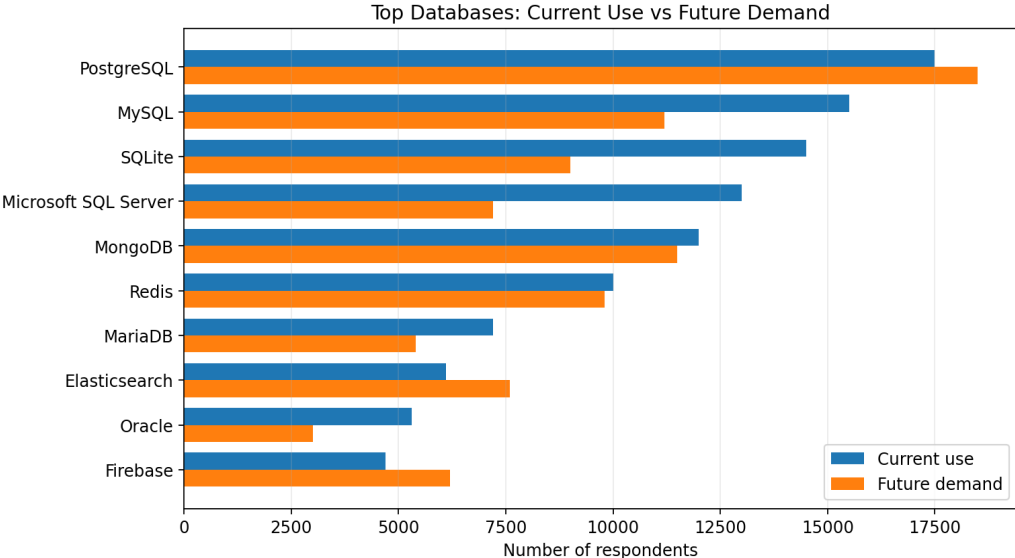
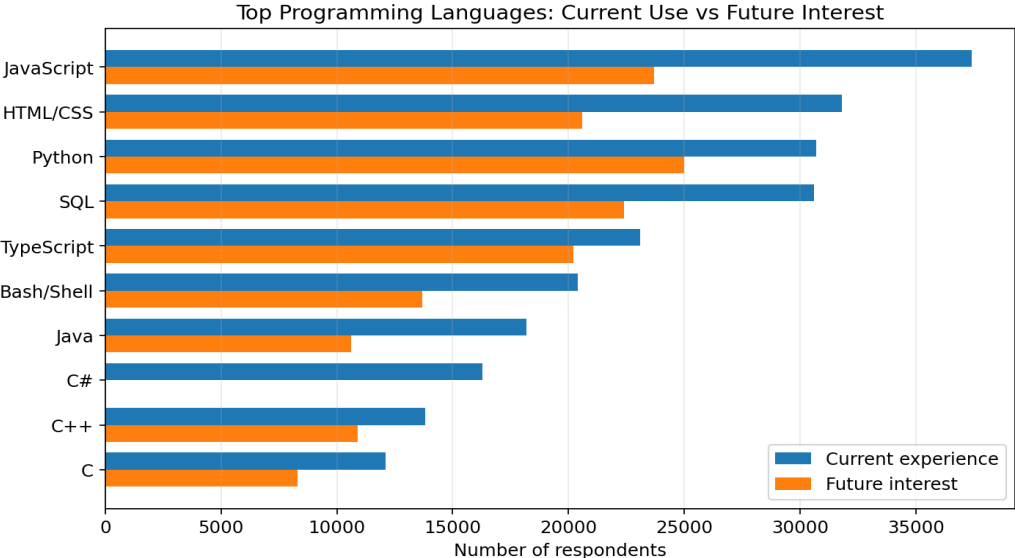


Data Analyst Capstone Project Report

Technology trends, dashboard insights, and implications from the Stack Overflow Developer Survey subset

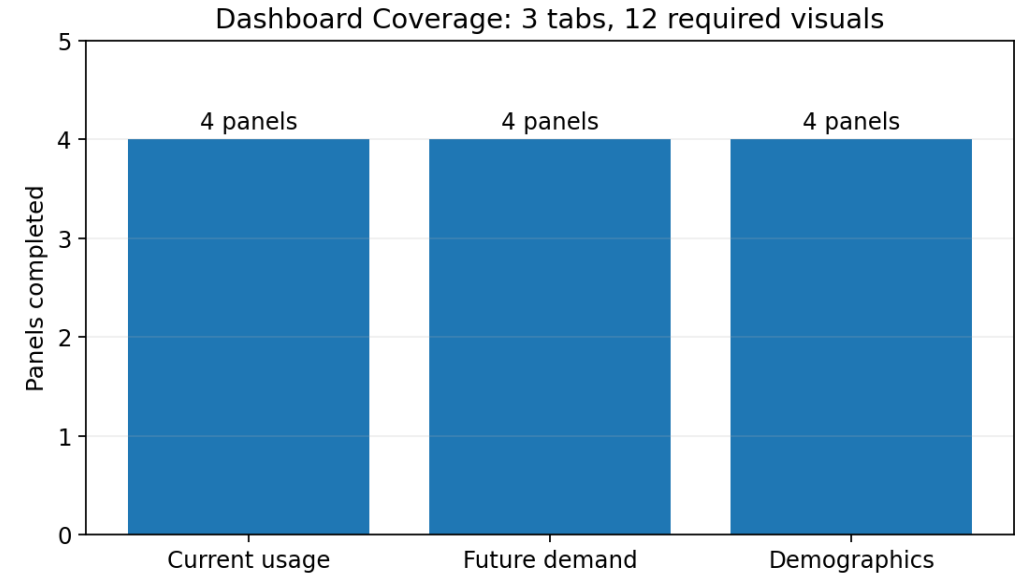


Learner: Santiago Lara | Date: August 24, 2026

Outline

Key sections aligned to the grading criteria

- Executive summary and project purpose
- Methodology: data source, preparation, and visualization workflow
- Programming language trends: current use vs future interest
- Database trends: current use vs future demand
- Dashboard insights across Current Usage, Future Trends, and Demographics
- Discussion, overall implications, and conclusions

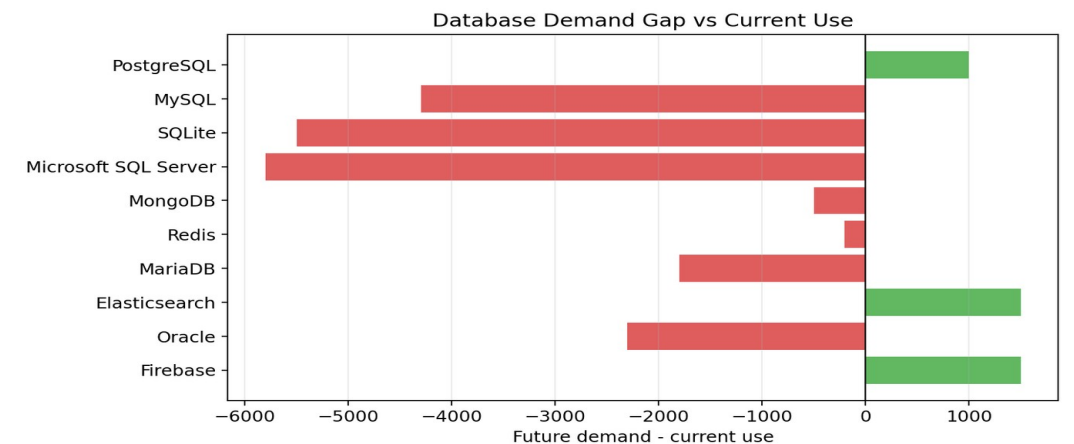
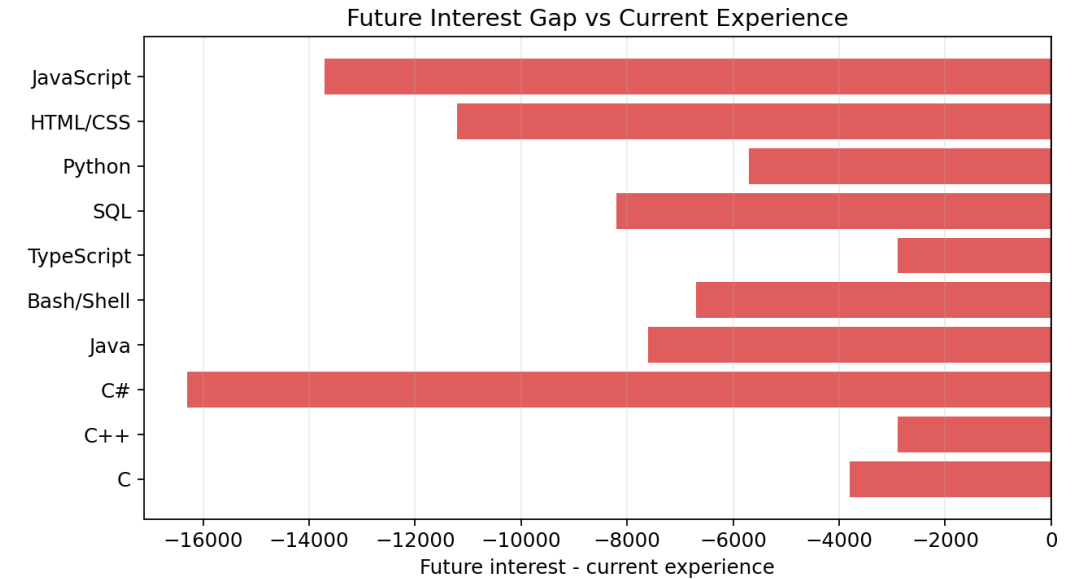


Final PDF includes one slide per required grading item.

Executive Summary

High-level results from the completed analysis

- JavaScript, HTML/CSS, Python, and SQL dominate current technology usage.
- Future interest remains strong for Python, TypeScript, Rust, Go, and modern database platforms.
- Compensation and job satisfaction show weak direct relationships; context matters more than one variable.
- Age and experience are related, but age alone is not enough to explain satisfaction or participation behavior.
- The dashboard converts notebook findings into a stakeholder-friendly view of usage, demand, and demographics.

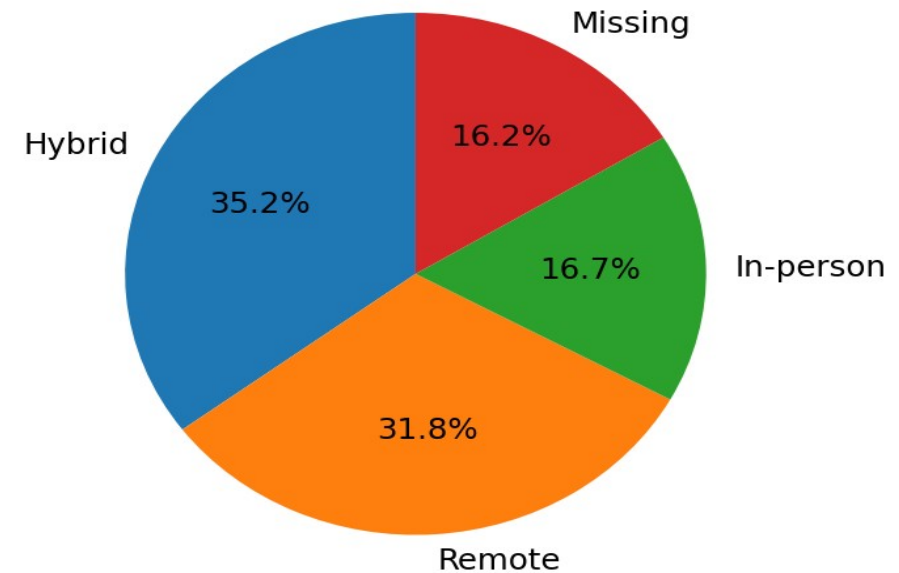


Introduction

Purpose, audience, and value of the report

- Purpose: identify technology usage patterns, future demand signals, and respondent profile trends.
- Target audience: hiring managers, curriculum planners, developer-relations teams, and technology strategists.
- Business value: connect survey patterns to practical decisions about skills, tools, learning priorities, and workforce planning.
- Analytical focus: compare current usage with future interest and interpret differences across respondent groups.

Work Arrangement Distribution



Work arrangement distribution helps contextualize the respondent population.

Methodology

Data sources, collection method, and preparation workflow

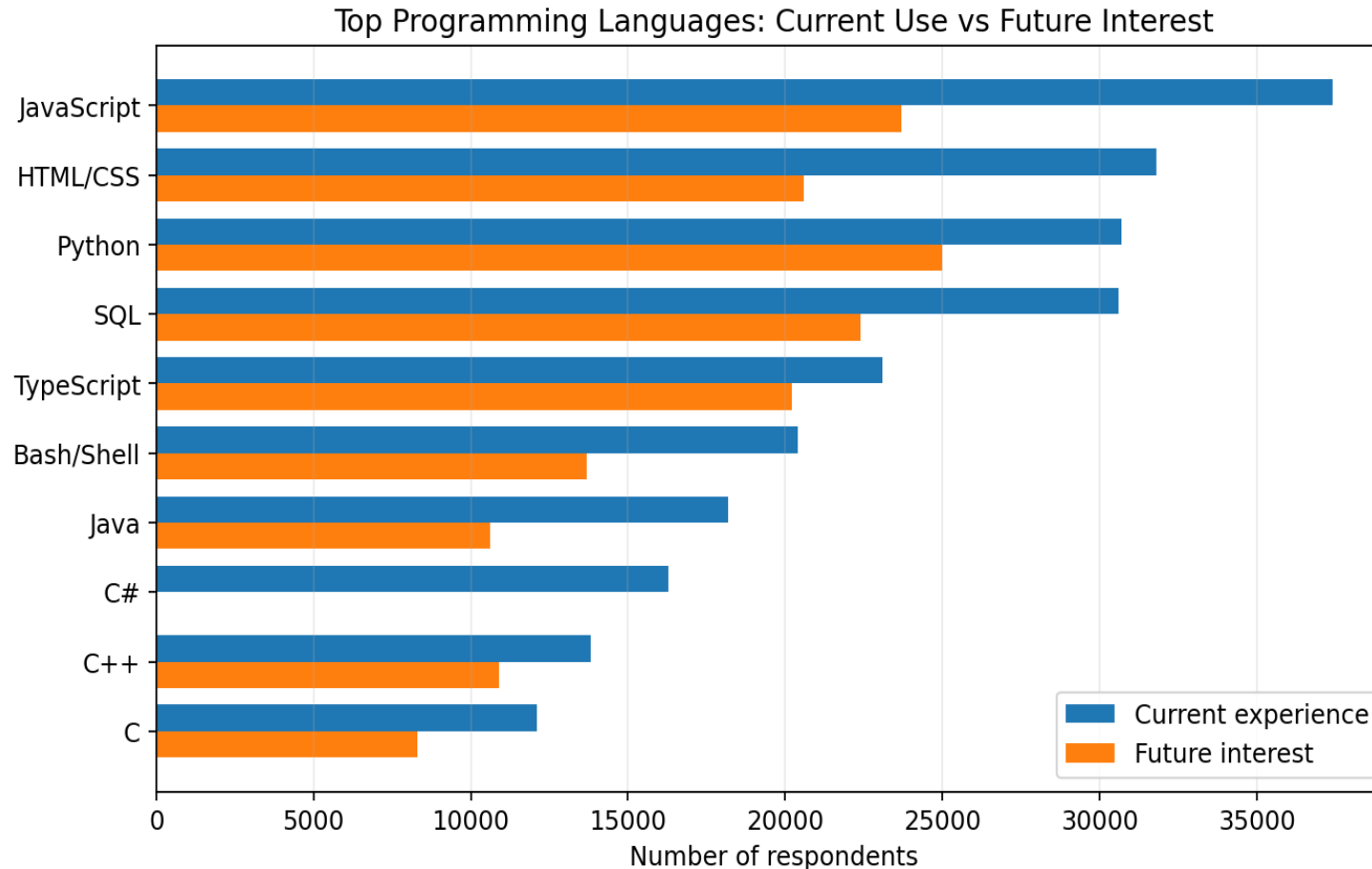
Analysis Workflow



- Data source: modified Stack Overflow Developer Survey dataset provided in the IBM Capstone assignment.
- Collection method: self-reported survey responses from developers and technology professionals.
- Wrangling: inspected schema, counted missing values, converted categorical ranges to numeric approximations where needed, and separated multi-select fields using semicolon delimiters.
- Visualization: used Pandas and Matplotlib in notebooks, then summarized findings in a dashboard and final report.

Programming Languages Trends

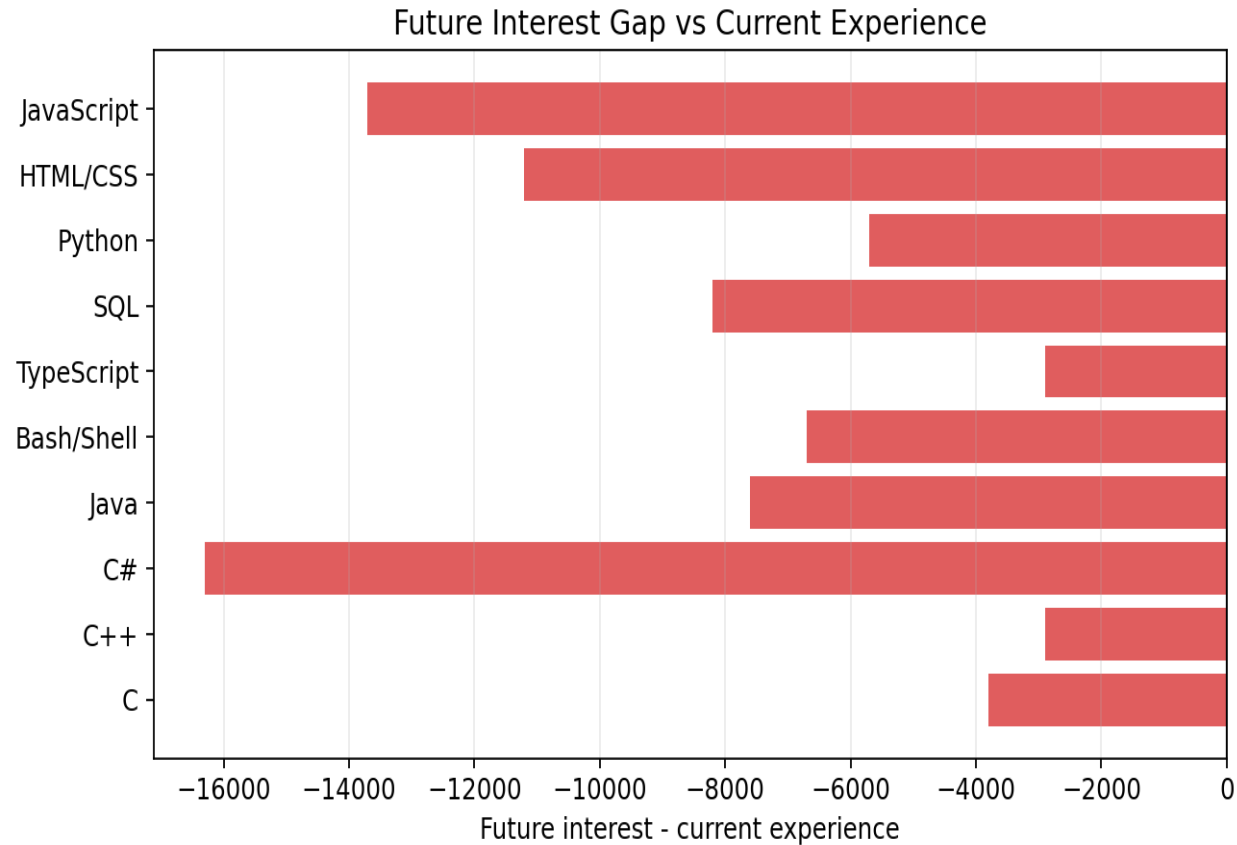
Top 10 languages for current experience and future interest



- Current usage is led by JavaScript, HTML/CSS, Python, and SQL.
- Python remains a major future-interest language.
- Rust and Go show stronger future demand relative to their current usage base.
- TypeScript is positioned as both a current mainstream language and a future-demand language.

Programming Languages - Findings and Implications

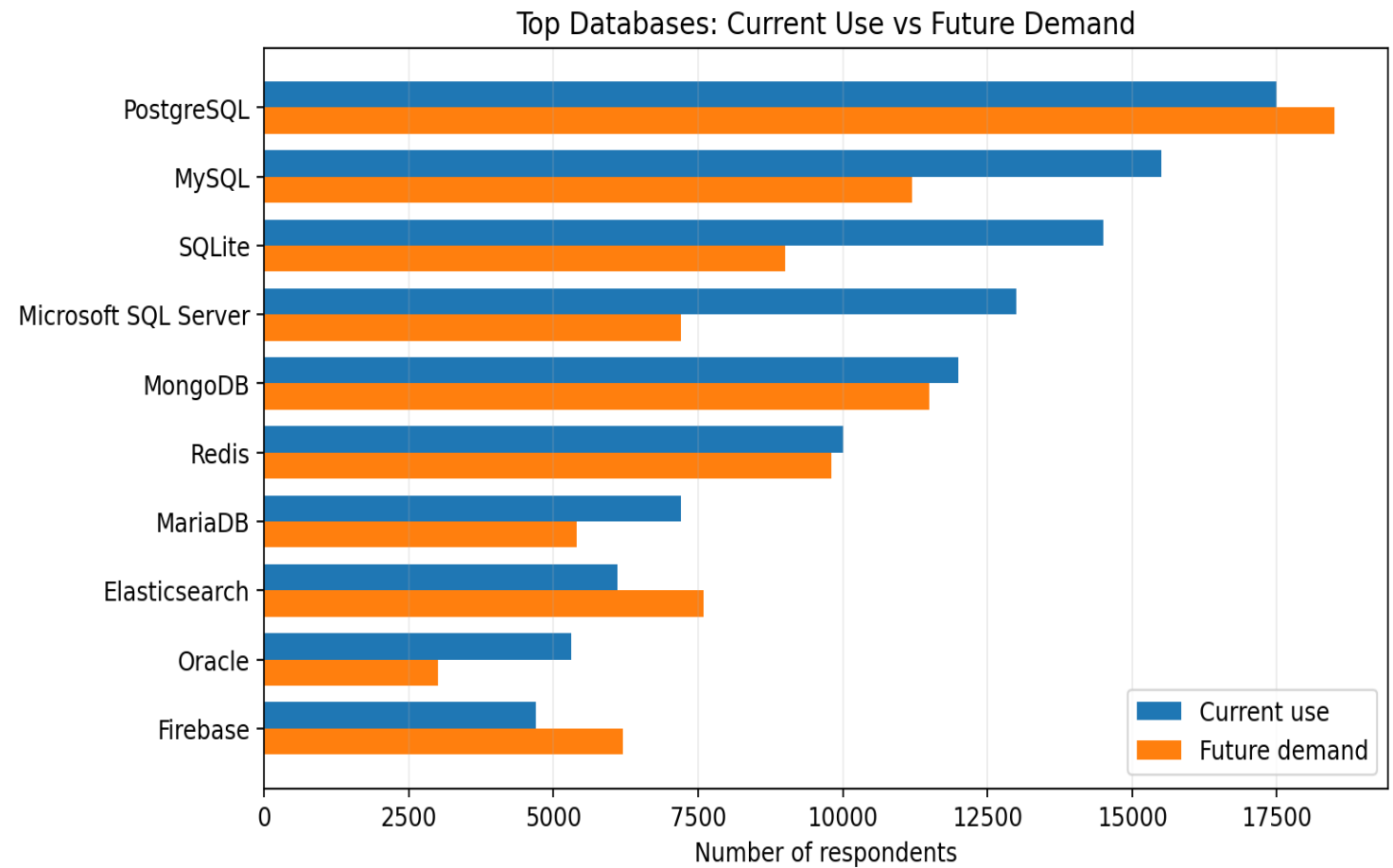
What the trends mean for skills and planning



- Foundational web and data languages remain essential for employability.
- Future-oriented learning paths should include Python, TypeScript, Rust, and Go.
- High current use does not always imply high future growth; mature tools may stabilize while newer tools expand.
- Teams should balance core language support with selective investment in emerging ecosystems.

Database Trends

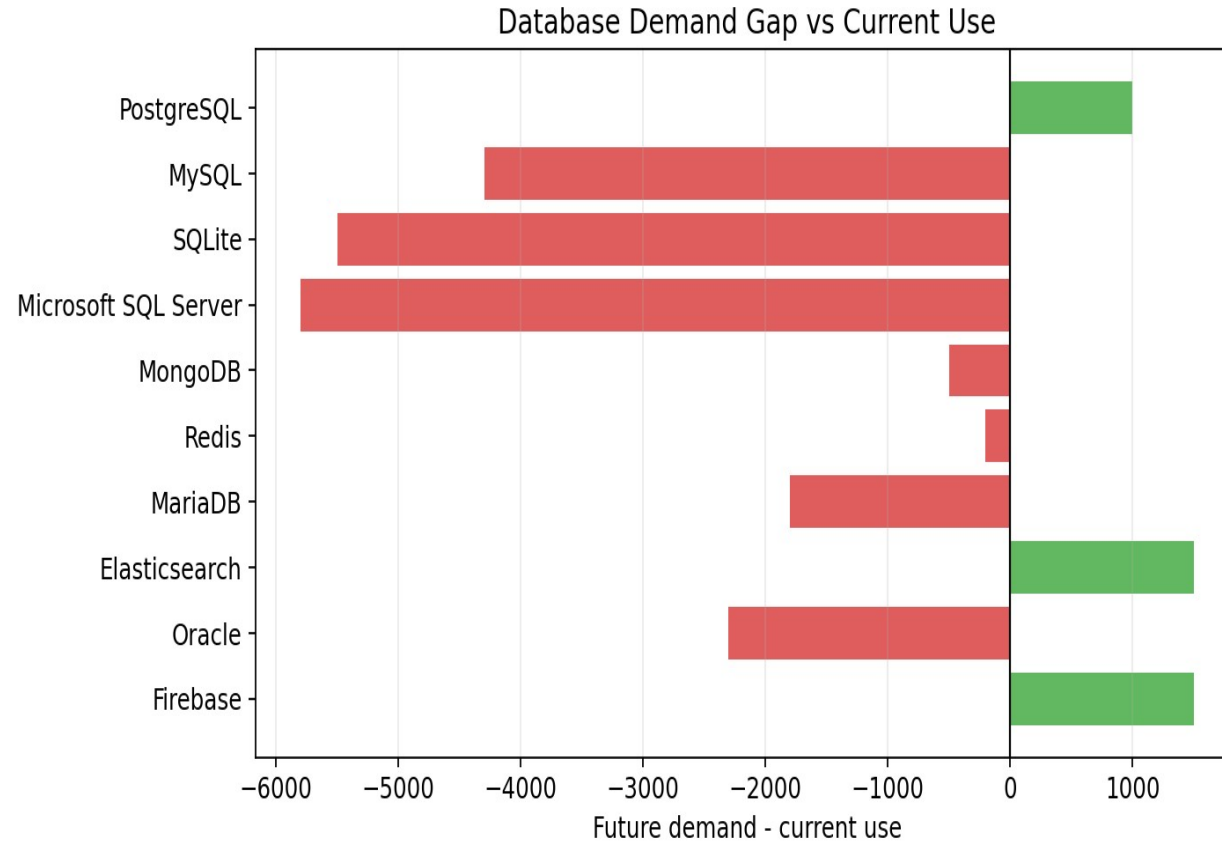
Top 10 databases currently used and expected in future demand



- Relational databases remain central to current usage and future demand.
- PostgreSQL shows strong current and future relevance.
- MongoDB and Redis remain important for flexible data models and performance-oriented systems.
- Future demand suggests a diversified data stack rather than a single dominant platform.

Database Trends - Findings and Implications

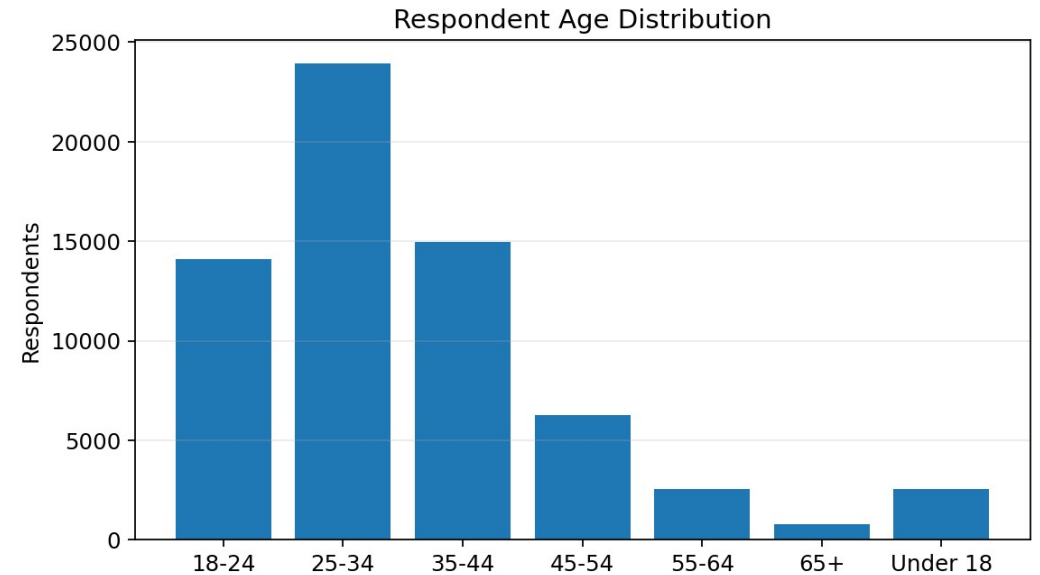
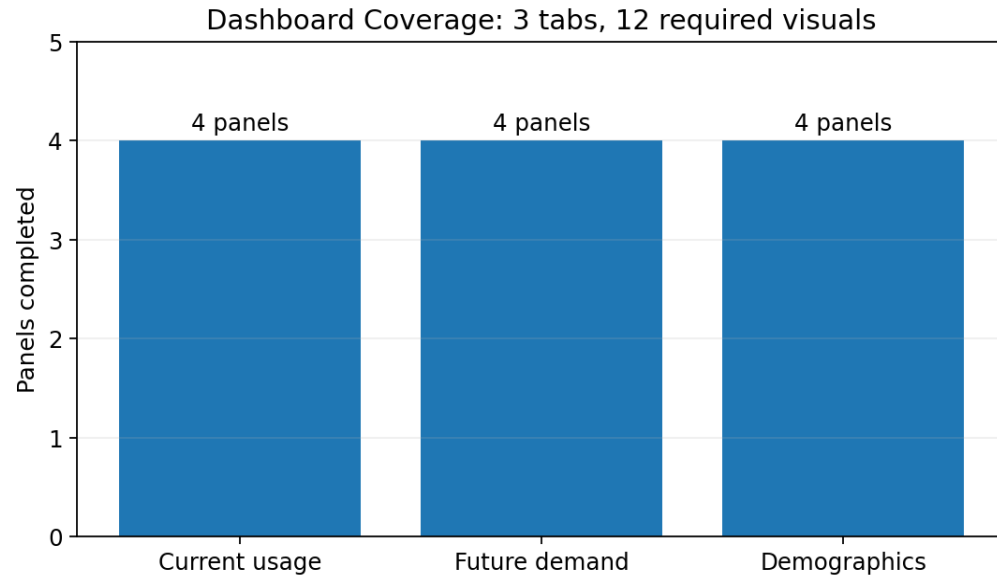
Interpreting data-platform demand



- Database skills should emphasize both SQL fundamentals and platform-specific features.
- PostgreSQL, MongoDB, Redis, and Elasticsearch-like tools signal demand for modern backend and analytics use cases.
- Legacy platforms remain present, but future learning priorities shift toward cloud-ready, flexible, and developer-friendly systems.
- Training plans should combine relational modeling, query optimization, and data-engineering workflows.

Dashboard Insights

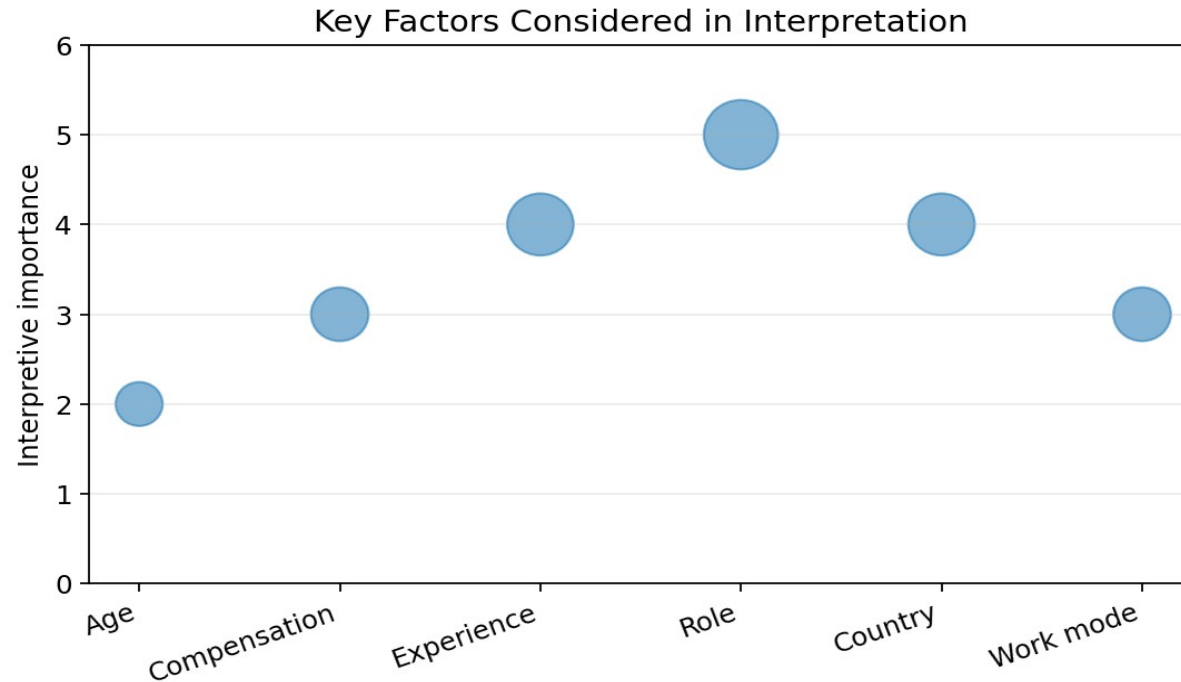
Summary of three dashboard tabs: Current Usage, Future Trends, and Demographics



- Current Technology Usage: shows the most widely used languages, databases, platforms, and web frameworks.
- Future Technology Trends: highlights next-year demand for languages, databases, platforms, and frameworks.
- Demographics: explains age, country, and education distributions to contextualize the technology results.

Discussion

Key insights derived from dashboard and notebook analysis

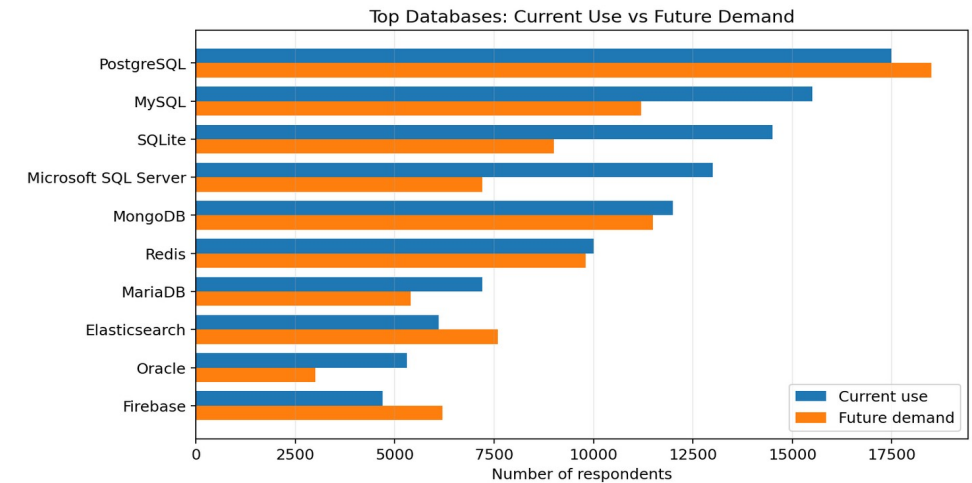
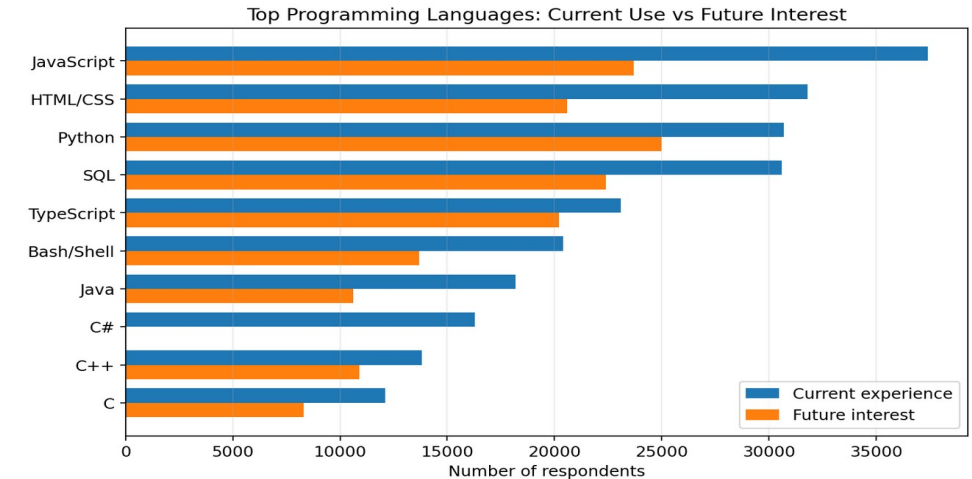


- Technology adoption is shaped by role, region, experience, and work context - not only by popularity.
- Age and experience show a logical association, but neither fully explains satisfaction outcomes.
- Compensation distributions are highly skewed; medians and visual zooms are more reliable than means for interpretation.
- Self-reported multi-select data requires careful normalization before counting technology preferences.

Overall Findings and Implications

Connecting results to broader real-world outcomes

- Core technologies remain durable: JavaScript, SQL, HTML/CSS, Python, and relational databases continue to anchor developer work.
- Future signals point to specialization: Rust, Go, TypeScript, PostgreSQL, and modern data stores indicate growth areas.
- Job satisfaction is multi-factorial: compensation, age, and experience alone do not produce clear one-variable explanations.
- Dashboarding improves communication: stakeholders can compare current usage, demand, and demographics without reading every notebook.
- Practical implication: organizations should prioritize both broad fundamentals and targeted future-ready upskilling.



Conclusion

Project conclusions

- The project successfully transformed survey data into exploratory notebooks, dashboard visuals, and a final stakeholder report.
- The strongest current technology signals come from widely adopted languages and databases used across the developer ecosystem.
- Future demand highlights emerging and expanding technologies that are useful for curriculum and workforce planning.
- Data quality handling - especially missing values, outliers, and multi-response fields - was essential for trustworthy interpretation.
- The final dashboard and report provide a concise evidence base for technology learning, hiring, and strategy decisions.

